



Data Analytics Capstone Topic: Monsoon & Macro Signals to Predict Stock Break-Out

Synopsis

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Formatting guide

1. The figures, diagrams, charts, and tables need to be properly labeled and referenced.
2. Grammatical correctness.
3. No spelling errors.
4. Unity in thought in each paragraph. Logical thoughts.
5. If using abbreviations, please define the full-form, then create the abbreviation, irrespective of the popularity of the same.



Introduction

Business Context :

India's summer monsoon powers farm income → rural spending/lower urban inflation → (Gross Domestic Product)GDP.

Markets often move when macro data surprise, yet most equity models ignore **weather**.

Gap:

Academic work links monsoon to GDP **or** looks at market re-balancing; **none** fuse rainfall + GDP/(Consumer Price Index)CPI/(Purchasing Managers Index)PMI + (repurchase rate)repo to forecast quarterly mid-cap returns.

Opportunity:

If a good monsoon plus supportive macro prints foreshadow a “break-out” quarter, investors can time exposure; policymakers can gauge climate sensitivity.



Scope and objectives

- i. **Research Question 1:** Can lagged monsoon rainfall anomaly and four macro indicators improve next-quarter excess-return forecasts for the (Indian index)NIFTY Midcap 100 index?
- i. **Research Question 2:** Do excess returns differ meaningfully between “good” and “poor” monsoon seasons?
- i. **Research Question 3:** Does rainfall lead next-quarter GDP growth, and can that inferred GDP improve market forecasts?
- i. **Research Question 4:** Does the (Reserve Bank of India)RBI’s policy stance (repo-rate change) amplify or dampen rainfall’s effect on next-quarter excess returns?



Scope and objectives

Sample size calculation

Question	Test / Model	Needed N	We Have
1 ElasticNet vs LightGBM (6 pred)	Rule $10 \times p \rightarrow 60$	62	✓
2 Good-vs-Poor rain t/ KS	$\geq 30/\text{group}$	34 good, 28 poor	~Adequate- $\rightarrow$ KS back-up
3 OLS Rain $\rightarrow$ GDP β	Power(0.80, $\alpha=.05$, $f^2=.15$) $\rightarrow 55$	62	✓
4 OLS interaction β_3 (rain $\times$ repo)	Same as RQ 3 (4 coeffs) $\rightarrow \geq 40$	62	✓



Data Description

Quarterly Data-Dictionary & Public URLs

Col / Feature	Description	Raw Source URL
date_q	Quarter-end date	derived
midcap_tr	NIFTY Midcap 100 Total-Return (daily → quarterly)	https://in.investing.com/indices/cnx-100-historical-data
nifty50_tr	NIFTY 50 Total-Return (daily → quarterly)	https://in.investing.com/indices/s-p-cnx-nifty-historical-data
rain_anom	All-India SW-Monsoon rainfall anomaly (%)	https://www.data.gov.in/catalog/rainfall-india
gdp_yoy	Real GDP YoY % (quarterly)	https://www.investing.com/economic-calendar/indian-gdp-quarterly-434
cpi_yoy	CPI headline YoY % (monthly → Q)	MOSPI release ¹
pmi_mfg	Manufacturing PMI (monthly → Q avg)	https://www.moneycontrol.com/economic-indicators/india-manufacturing-pmi-1414724
pmi_comp	Composite PMI (monthly → Q avg)	https://www.moneycontrol.com/economic-calendar/india-markit-composite-pmi/2323490
repo_chg	RBI repo-rate Δ (bps) within Q	https://masterstrokeonline.com/download-tool/download/34/RBI%20Repo%20Rate%20History.pdf
ret_prev_q	Lagged Midcap return	calculated
excess_next_q	Target: (Midcap – Nifty50) return in Q + 1	calculated
rain_repo_int	Interaction = rain_anom × repo_chg	engineered
gdp_pred_from_rain	OLS-implied GDP from rain	engineered



Data Description

GitHub-repo:

<https://github.com/Workerhash/capstone>

-Contains raw files, initial notebook.



Analytic approach

Research question 1:

Predictive Boost : Will rain + macro lags beat lag-return baseline for **next-Quarter excess return**?

Proposed Solution:

Baseline **ElasticNet**(ret_prev_q) vs Enriched **LightGBM**(ret_prev_q + 5 lags).

5-fold walk-forward CV → compare R^2 , MAE.

Pass: $\Delta R^2 \geq 0.10$ & MAE < 10 %.



Analytic approach

Research question 2:

Monsoon Effect: Do excess returns differ in **good vs poor** rain years?

Proposed Solution:

- Classify quarters by IMD anomaly; perform two-sample **t-test** & **KS**;
- boxplot.



Analytic approach

Research question 3:

Exploring Rain→GDP Link: Does rain predict **GDP t+1**, and does adding inferred GDP lift model?

Proposed Solution:

- OLS: $GDP_{t+1} = \alpha + \beta \cdot Rain_t$ (checking if we have significant β).
- Create gdp_pred_from_rain; retrain LightGBM;
- success if $R^2 \geq 0.03$ or $MAE \geq 5 \%$.



Analytic approach

Research question 4:

Driver Attribution: Which factor block (rain vs each macro) drives accuracy?

Proposed Solution:

- Permutation importance / drop-one** on LightGBM;
- rank ΔR^2 per feature.
- Produces “heat bar” so business can see key levers.



Analytic Approach

Step	Action	Code hint
1	Create interaction term: $rain_repo_int = rain_anom_lag * repo_chg_lag$	<code>df['rain_repo_int'] = df['rain_anom_lag'] * df['repo_chg_lag']</code>
2	Fit OLS: $excess_next_q = \alpha + \beta_1 \cdot rain + \beta_2 \cdot repo + \beta_3 \cdot rain * repo + \varepsilon$	<code>sm.OLS(y, sm.add_constant(X)).fit()</code>
3	Test β_3 (interaction) – significant if $p < 0.05$.	<code>results.pvalues['rain_repo_int']</code>
4	If significant, add $rain_repo_int$ to LightGBM feature list and re-run walk-forward CV.	<code>update feat_cols</code>
5	Success rule: interaction model lifts $R^2 \geq 0.02$ or MAE $\geq 3\%$ vs previous enriched model.	<code>compare metrics</code>



Evaluation Metrics

Metric	Formula	Used In
R^2	$1 - \Sigma(\hat{y}-y)^2 / \Sigma(y-\bar{y})^2$	RQ-1, 3, 4
MAE	Σ	$\hat{y}-y$
Cohen d	$(\mu_1-\mu_2) / \sigma_{\text{pooled}}$	RQ-2
KS-D	max	$F_{\text{good}} - F_{\text{poor}}$
β , p-value	from OLS	RQ-3, 4



Recommendation and applications

Fund managers: overlay “Monsoon + Macro” signal on existing factor models to time mid-cap exposure.

Sell-side research: quarterly *Monsoon Tracker* note converting new rainfall data into market calls.

Policy & ESG analysts: quantify climate → GDP → equity channel to inform agri-policy and climate-risk disclosure.

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Thank You